Mind–Brain Type Identity Theory

Smart's identity thesis, its advantages, and the objections he answers

This workbook consolidates what you have covered in the slideshow and handout. Work through it in order, or jump to the part you need. Every task can be done on your own, so if you missed a lesson you can still catch up. Answers to the recall tasks are at the back: try first, then check.

Name: _____ Date started: _____

In this workbook

1. Where am I now? A self-check
2. Key terms
3. Quick recall
4. Argument reconstruction
5. Objections and evaluation
6. Exam practice (3, 5, 12 and 25 marks)
7. Answers

> **Key Thesis**
> All mental states are **identical** to brain states. This is an **ontological** reduction (about what exists), not an **analytic** one: 'mental state' and 'brain state' do not mean the same thing, just as 'lightning' does not *mean* 'electrical discharge', though they are the same thing.

Part 01 Where am I now?

How to use this section

Tick a column for each statement **before** you start, then again **when you finish**. Green = I could explain it to someone else. Amber = I half get it. Red = not yet.

I can...	Start	End
state the identity thesis precisely	☐ R ☐ A ☐ G	☐ R ☐ A ☐ G
explain ontological versus analytic reduction	☐ R ☐ A ☐ G	☐ R ☐ A ☐ G
explain type versus token identity	☐ R ☐ A ☐ G	☐ R ☐ A ☐ G
explain why the theory solves the problem of causation	☐ R ☐ A ☐ G	☐ R ☐ A ☐ G
explain and answer Smart's objections (the peasant, location, privacy)	☐ R ☐ A ☐ G	☐ R ☐ A ☐ G
explain multiple realisability and why it is the main problem	☐ R ☐ A ☐ G	☐ R ☐ A ☐ G
reach and defend a judgement on type identity theory	☐ R ☐ A ☐ G	☐ R ☐ A ☐ G

Part 02 Key terms

Task 1 · Match the term to its definition

Write the letter of the correct definition next to each term.

Term	Letter
Physicalism	
Type identity	
Token identity	
Ontological reduction	
Analytic reduction	
Strict identity	
A posteriori	
Multiple realisability	

	Definition
A	The same mental state can be realised by different physical states in different beings.
B	Showing that one term means the same as another, by analysing concepts.
C	Known through experience or empirical investigation, not by reason alone.
D	Each particular mental event is identical to some particular brain event, though types may differ.
E	Showing that one kind of thing is nothing over and above another kind of thing.
F	The 'is' of identity: X and Y are one and the same thing, not merely correlated.
G	Everything is physical or supervenes on the physical.
H	Each type of mental state (e.g. pain) is identical to a type of brain state.

Task 2 · Fill the gaps

Use the word bank. Each word is used once.

Key idea Word bank: H_2O · posteriori · brain · discharge · realisability · ontological · causation · identity

Smart argues that sensations are nothing over and above _____ processes. He uses 'is' in the sense of strict _____, as in 'lightning is an electrical _____'. The theory is an _____ reduction, not an analytic one, because the identity is discovered a _____, like water and _____. Its main advantage is that it avoids the problem of _____. Its main problem is multiple _____: octopuses and other animals may feel pain with very different brains.

Task 3 · In your own words

Using the morning star and evening star, explain how two terms can refer to the same thing even though they do not mean the same thing.

Part 03 Quick recall

How to use this section

Answer from memory first. Then check your notes, and add anything you missed in a different colour. That second colour shows you exactly what to revise.

1. What does ‘nothing over and above’ mean in Smart’s thesis?

2. Distinguish type and token identity.

3. Why does identity theory solve the problem of causation?

4. State the ‘illiterate peasant’ objection.

5. How does Smart answer it?

6. What is Smart’s reply to ‘after-images are not in physical space’?

7. Why is ‘I can imagine sensations without brain processes’ not decisive?

8. State the multiple realisability objection.

Part 04

Argument reconstruction

> **How to use this section**
> Each argument below has been jumbled. Number the lines into the right order (premises first, conclusion last). Then answer the questions underneath. Being able to set out an argument in standard form is exactly what the 5-mark and 12-mark questions reward.

The case for type identity theory

> *'When I say that a sensation is a brain process or that lightning is an electrical discharge, I am using 'is' in the sense of strict identity'*
>
> — J.J.C. Smart, *Sensations and Brain Processes (1959)*

Order	Jumbled line
	Mental states have physical causes and physical effects.
	Therefore, mental states are identical to brain states.
	Science has found close correlations between mental states and brain states.
	Physics explains every physical event by physical causes alone.
	The simplest explanation of these correlations is that mental states are brain states (Ockham's razor).

→ Which premise appeals to simplicity? Is simplicity a good reason to believe something is true?

→ Why does premise 2 make dualism look implausible?

The multiple realisability objection

Order	Jumbled line
	Then only creatures with that brain state can feel pain.
	Therefore, pain is not identical to one specific type of brain state: type identity is false.
	If type identity is true, pain is identical to one specific type of brain state.

Order	Jumbled line
	Octopuses, other animals and possibly aliens feel pain with very different brains.

→ Could the identity theorist narrow the claim to 'human pain = brain state B'? What would be lost?

→ Does token identity escape the objection? What does it give up?

Part 05 Objections and evaluation

How to use this section

For each objection: explain the problem in your own words, write the best response, then judge whether the response works. The hint column is there if you are stuck; cover it up if you want a harder challenge.

Objection	Hint	The problem, in my own words	Best response	Does it work? Why?
The illiterate peasant	Talk of pains without any neuroscience.			
Science might change	What if future science shows sensations aren't brain processes?			
Location	After-images aren't in physical space; brain processes are.			

Objection	Hint	The problem, in my own words	Best response	Does it work? Why?
Privacy	Sensations are private; brain processes are public.			
Conceivability (dualist argument)	I can imagine pain with no brain process.			
Multiple realisability	Octopus pain, alien pain.			
Qualia / hard problem	What it is like to see red.			

Ranking the objections

Which objection is the **most** damaging, and which is the **least**? Justify your choice: a strong objection attacks a key premise and is hard to answer.

Most damaging: _____ because...

Least damaging: _____ because...

Building a judgement

Are mental states identical to brain states? Weigh the theory's strength on causation against the problem of multiple realisability.

Key idea Sentence starters: *The strongest argument for this view is... because...* · *However, this is undermined by...* · *A defender might reply that... but this fails / succeeds because...* · *Therefore, on balance...*

Part 06 Exam practice

How to use this section

On Paper 2 the **3-, 5- and 12-mark** questions test **AO1** only: knowledge and understanding, shown through precise, accurate explanation. The **25-mark** question is **5 marks AO1 + 20 marks AO2**: you must argue, evaluate and reach a judgement that your essay supports throughout.

Spend roughly a minute per mark. Plan before you write.

3 marks · What is meant by 'multiple realisability'?

Key idea AO1 · A precise definition or short point. One or two accurate sentences; no evaluation needed.

5 marks · Explain the difference between ontological and analytic reduction, with reference to mind–brain type identity theory.

Key idea AO1 · Set out the idea or argument clearly and in logical order. Precise terminology; a standard-form layout often works best.

Planning frame

1. Define analytic reduction (meaning/synonymy).
2. Define ontological reduction (what exists).

3. Identity theory is ontological: 'pain' ≠ 'brain state B' in meaning.
4. Illustrate: water/H_2O or lightning/electrical discharge.

12 marks · Explain how the multiple realisability of mental states challenges mind–brain type identity theory.

Key idea AO1 · A full, precise explanation. Show how the ideas connect logically, step by step. Do **not** evaluate: explain.

Planning frame

1. State type identity theory precisely.
2. Explain multiple realisability with examples (octopus, aliens).
3. Show step by step why it contradicts *type* identity.
4. Explain how this pushes towards token identity or functionalism.

25 marks · Is mind–brain type identity theory true?

Key idea AO1 (5) + AO2 (20) · State your line of argument in the introduction and keep to it. Every paragraph should argue, not just describe: point → objection → response → mini-judgement. End with a conclusion that follows from what you have argued.

Model structure

Section	What to do	My plan
Introduction	Define the key terms. State your thesis: the answer you will defend.	
Paragraph 1	Advantage: solves the problem of causation, Ockham's razor → challenge → mini-judgement.	
Paragraph 2	Smart's objections (peasant, location, privacy) and his replies → mini-judgement.	
Paragraph 3	Dualist arguments (conceivability, knowledge argument) → conceivable ≠ possible → mini-judgement.	
Paragraph 4	Multiple realisability → narrowing or token identity → mini-judgement.	
Conclusion	Restate your judgement and explain **why** the balance of argument supports it. No new arguments.	

Write your essay on lined paper, or continue below.

Reflect

Go back to Part 01 and fill in the **End** column. Which statement is still red or amber? Write one specific thing you will do about it.

Part 07 Answers

> **How to use this section**
> Use these to check Parts 02–04. Exam answers are not given here: bring them to your teacher, or mark against the AO guidance.

Task 1 · Matching

Physicalism = G · Type identity = H · Token identity = D · Ontological reduction = E · Analytic reduction = B · Strict identity = F · A posteriori = C · Multiple realisability = A

Task 2 · Gaps

(1) brain · (2) identity · (3) discharge · (4) ontological · (5) posteriori · (6) H_2O · (7) causation · (8) realisability

Quick recall

1. Sensations are not extra things in addition to brain processes; they just are those processes (as a nation is nothing over and above its citizens).
2. Type: each kind of mental state = a kind of brain state. Token: each individual mental event = some individual brain event.
3. If mental states are brain states, mental causes are physical causes, so no non-physical cause is needed; consistent with physics.
4. A person can talk about their after-images and pains while knowing nothing about neurophysiology, so sensation talk is not brain talk.
5. Two terms can refer to one thing without the speaker knowing (morning star/evening star). The identity is not about meaning.
6. He identifies the *experience* of an after-image with a brain process, not the after-image itself.
7. Conceivability shows only that the identity isn't known a priori; like water = H_2O, it can be necessary but known a posteriori.
8. Animals with very different brains feel pain; so pain cannot be identical to one type of human brain state.

Arguments in order

The case for type identity theory

1. Mental states have physical causes and physical effects.
2. Physics explains every physical event by physical causes alone.
3. Science has found close correlations between mental states and brain states.
4. The simplest explanation of these correlations is that mental states are brain states (Ockham's razor).
5. Therefore, mental states are identical to brain states.

The multiple realisability objection

1. If type identity is true, pain is identical to one specific type of brain state.
2. Then only creatures with that brain state can feel pain.
3. Octopuses, other animals and possibly aliens feel pain with very different brains.
4. Therefore, pain is not identical to one specific type of brain state: type identity is false.

Objections: model responses

Objection	Possible response
The illiterate peasant	Smart: reference without knowledge, like morning star/evening star; the identity is discovered, not analytic.
Science might change	Smart: this only shows 'I have an after-image' doesn't *mean* a brain-process statement; the same is true of lightning.
Location	Smart: the *experience* of an after-image is the brain process; the after-image itself is not a thing.
Privacy	Future neuroscience may give equal access; privacy may be epistemic, not ontological.
Conceivability (dualist argument)	Conceivable ≠ possible; the identity is necessary a posteriori, like water = H_2O.
Multiple realisability	Narrow the identities to species, or retreat to token identity; functionalism is the usual next step.
Qualia / hard problem	The identity theorist says the quale just is the brain state; critics say this leaves the felt quality unexplained.